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MindTrace-ID Personalized Cognitive Drift Modeling for Early Mental Health Risk Detection Using Longitudinal Clinical Interviews

Abstract—Early detection of mental health decline continues to be a challenge owing to significant inter-individual differences, delayed toxicity profile and the reliance on population-based emotional cues by current artificial intelligence models with limited clinical interpretability. To overcome these drawbacks, in this paper we consider MindTrace-ID, a personalized mental health detection framework capturing cognitive drift with longitudinal micro-behavioral and textual signals rather than the static sentiment ones. This method builds a personalized cognitive baseline for each subject and dynamically scores deviations in terms of syntactic complexity, semantic coherence, interaction rhythm, as well as response latency. These disparate indicators are adjudicated using a drift-aware adaptive inference strategy that prioritizes temporal deviation trends over absolute emotional polarity. To provide transparency and clinical relevance, MindTrace-ID incorporates an interpretable reasoning level which encodes detected drifts in behavior to cognitive markers (e.g., attentional instability, cognitive slowing). The framework performance is tested on the publicly available DAIC-WOZ data base of longitudinal records of interviews with clinical annotations for depression assessment. Results show that our proposed detector obtains a global detection accuracy of 91.6%, which significantly outperforms deep learning baselines trained on population data by 7.8% and simultaneously improved the early-stage risk detection-recommend recall from 79.2% to 88.4%. There are increased benefits for personalized baseline modeling which also decreases false positive rates by 32.5%, and ablation analysis shows that drift cognitive modeling yields a performance increase of 19.3% when compared to static representations. These results demonstrate that personal cognitive drift analysis is a reliable, interpretable, and scalable approach for early clinical intervention in proactive monitoring of mental health.

Keywords: Mental health detection; Cognitive drift analysis; Personalized baseline modeling; Explainable artificial intelligence; Longitudinal behavioral analysis; Depression risk assessment

1 Introduction

Mental health diseases, such as depression and anxiety, are growing worldwide however detection often comes late due to gradual emergence of symptoms and high inter-patient variation [1][2]. This diagnostic assessment is common to depend on the one-time clinical assessments or self-reporting questionnaires [3], and it suffers from

subjective bias, less frequent monitoring, as well as late detection of symptoms. Such methods are likely to miss the protracted behavioural and cognitive changes that precede clinical diagnosis by many years [4][5]. Recent AI solutions demonstrate some promise on automating MH assessment, yet most existing models capture population-level emotional patterns and fail to be sensitive to individual behaviour trajectory. Consequently, they are subject to misclassification and delayed risk recognition. Most importantly, mental health decay is indicated in subliminal and gradual cognitive anomalies rather than immediate emotional transformations, therefore demanding kinematic aware and personalized detection mechanisms.

1.1 Gaps in Existing AI Approaches

The latest AI based mental health detection techniques have been mainly based on sentiment analysis, emotional polarity and static text representations from single interactions. These methods are effective in recognizing emergency cases of emotional distress [6][7] but have low performance for the early cases where signals are weak or inconsistent. Static embeddings also overlook longitudinal dependencies by which a temporary emotional stage can be differentiated from the persistent cognitive decline. Furthermore, most DL models are used as black-box systems with limited interpretability and clinical understanding. As individual baseline modeling is not available here meaningful deviation analysis cannot be performed, since changes in behaviour are compared to norms and not an individual historical standard. Therefore, current methods do not have drift-informed reasoning and clinical interpretable outputs needed for robust early mental health monitoring [8].

1.2 Core Idea of MindTrace-ID

In order to address such issues, we present a personalized mental health detection framework referred as MindTrace-ID that learns the cognitive drift along longitudinal clinical interviews [9]. It formulates a personalized cognitive baseline for each user and dynamically measures distortions in various micro-behavioral and linguistic parameters, including syntactic complexity, semantic coherence, coupling pace, latency-of-response. MindTrace-ID captures dynamic changes in cognition that are reflective of early stage risk, by focusing on the temporal pattern of aberrant emotional state rather than the absolute emotional state. An explainable reasoning layer additionally translates detected drift patterns to interpretable cognitive markers, offering a transparent and clinically relevant evaluation.

1.3 Contributions

The contribution of this paper is summarized as:

- A novel cognitive drift modeling framework personalized to early mental health risk estimation is presented, using individualized baselines and

longitudinal behavioral analytics rather than population-based sentiment mechanisms.

- A multi-index longitudinal feature representation and a drift-aware adaptive inference module is presented to model and prioritise sustainable syntactic, semantic, temporal, and interaction-level behavioral changes.
- An explainable reasoning layer is introduced to translate the detected cognitive drift patterns into clinically interpretable markers, and comprehensive testing on the DAIC-WOZ dataset shows that our model can achieve better accuracy, early-stage recall and lower false positive rate.

The rest of the paper is structured as follows. Related work and the research gap that justifies the proposed approach are studied in Section 2, while Section 3 details the dataset and preprocessing steps. Section 4 introduces the MindTrace-ID approach, Sections 5 and 6 describe experimental setup and results analysis, while we finish with discussion and future work in Section 7.

2 Literature Review

Ding et al. (2025) evaluated the decision-point trade-offs of machine learning and deep learning in mental health disorder recognition from social media text. They compared the traditional machine learning methods (logistic regression, random forest and LightGBM) with deep learning architectures like ALBERT and GRU networks. The authors showed that although both models reached similar performance on medium-sized datasets, machine learning models could provide better interpretability (feature importance) while deep learning ones were more robust to complex linguistic patterns. The study also pointed out that model selection should trade-off between performance, interpretability and dataset size and computation efficiency, stressing the on-going tradeoff of explaining patterns mental health detection systems [1]. Tiwari et al. (2024) studied stress detection from EEG signals and they have reviewed a full list of signal processing method, feature extraction technique and machine learning approach used in this field. Their work highlighted the need for non-invasive and real-time monitoring of physiology parameters to detect stress at an early stage, and addressed issues such as signal noise, inter-subject difference, and system scalability. While they demonstrated efficacy, the approaches described above depended on specialized hardware, and did not consider behavior or language context, thus were unsuitable for large scale cheap monitoring of mental health [2]. Matthews et al. (2022) studied loneliness as a general risk factor of mental health problems based on large scaled longitudinal cohort samples in the United Kingdom and New Zealand. Using the UCLA Loneliness Scale, we showed that loneliness can effectively classify individuals at-risk for depression, anxiety and self-harm providing classification accuracy comparable to well established

sleep-quality measures. Although clinically relevant, the method depended on questionnaire-based screening and lack of time-varying behavior change susceptibilities highlight a need for continuous and personalized diagnostic frameworks [3]. A deep learning-based proof-of-concept system for relapse detection and estimation of depression based on audiovisual channels from clinical interview videos was suggested by Othmani and Zeghina (2022). Their similarity-based anomaly-detection method compared audiovisual patterns extracted from depression-free and depressed states, thereby allowing us to predict relapse without any user involvement. On the DAIC-WOZ dataset we obtained good accuracy, but instead of individual-based baseline drift modeling, matching for similarity was emphasized and little explanation as to what changed in terms of underlying cognition [4]. Wang et al. (2024) investigated on-device artificial intelligence for depression detection using wearable internet of things. Their work underscored the benefits of leveraging psychophysiological data, such as sleep and activity, in continuous monitoring of mental health and showed better performance with ensemble learning. Even if the results are promising, wearable-based solutions rely on sensor availability and adherence, and they usually fail to take into account the linguistic and interaction-level behavioral cues that can be found in clinical interviews [5]. Hinduja et al. (2022) leveraged social media users as distributed social sensors and presented a proactive discord monitor system through Twitter data. They used LSTM-based models to identify individuals at risk in the early stage combined with intensive text preprocessing and sentiment analysis. Although successful in monitoring at the population level, we employed population level patterns and personal temporal baselines were absent which did not show its full potential for personalized early intervention [6]. Sharma et al. (2023) developed a multimodal emotional detection based on text, audio and video signals, relying on transfer learning and decision-level fusion. Their EmoHD model achieved state-of-the-art results on the benchmark datasets by aggregating heterogeneous affective cues. However, the model was for discrete emotion recognition but not continuous cognitive change over time, and no personalized drift analysis was made for early risk-detection of mental health [7].

Lin et al. (2024) were able to perform a large meta-analysis for the detection of mental health problems in sexual minorities on mainland China. Their results exposed wide variance in detection rates as a function of measurement instruments, cut-offs, and demographic moderators. The study stressed that context-aware and population-sensitive methods of assessment were essential by implication, the inadequacies of one-size-fits-all detection strategies [8]. Sharma et al. (2022) proposed an EEG-based stress detection method based on the entropy features and an evolutionary-optimized support vector machine. Providing high accuracy on the

short-time EEG signals, the paper showed that it is feasible to detect stress quickly. However, dependence on physiological indexes and hardware-based configuration limited the scalability, and could not mitigate behavioral or linguistic expression of deteriorating mental health [9]. Lin et al. (2023) combined external psychological knowledge to social-network-based mental health detection models by including diagnostic criteria and mild screening instruments. Their knowledge-sensitive model enhanced classification performance and reduced the dependence on data-driven decision forums. Although this methodology improved interpretation, it did not explicitly model longitudinal behavioral drift or individual baselines [10]. Yang et al. (2022) proposed tough interpretable mental health analysis as a text generation task and founded instruction-tuned large language models. Their MentaLLaMA network provided human-level explanations in addition to predictions, which help address the interpretability problem. However, it depended on the enormous first-step training data and was mainly based on social media text but not historical clinical engagement [11]. Nawindah and Wulandari (2023) have designed a mobile-based self-screening application as an attempt in order to assist the adolescent mental health assessment. Their app, Smart-Senyum also enhanced awareness and early recognition of mental health conditions in young people. The system was however based on self-reported questionnaires and did not have continuous behavior monitoring features [12]. Olawade et al. (2024) conducted an extensive analysis of the applications of AI-based technologies in mental health care, with a focus on early detection, personalized medicine and ethical aspects. The importance of privacy, bias and transparency in the application of AI to mental health were emphasised as key challenges that need to be addressed when designing explainable, responsible AI systems for mental health applications [13]. Tutun et al. (2023) developed an AI-enabled DSM-5 decision support tool, which simplified classical mental health assessment tools through reducing questionnaire length and enhancing diagnostic accuracy. This approach successfully identified candidates for removal; however, its design was focused on the questionnaire and it did not take advantage of longitudinal behavioural or linguistic signals [14]. Hybrid deep learning architecture based on textual and audio features has applied in depression detection by Marriwala and Chaudhary (2023) on DAIC-WOZ dataset. Their results showed that this approach works well in the case of audio-based CNNs. However, the model primarily focused on predictive accuracy rather than interpretability and did not include any explicit approach to model personalized temporal cognitive drift [15].

2.1 Research Gap

While earlier works have investigated textual, physiological and multimodal techniques in machine learning and deep learning models for mental health detection, there are several important open questions remain. Current methods are largely based

on aggregated sentiment across the population or static feature representations, which can have limited capacity to describe individual difference and subtle cognitive shifts that precede manifest symptoms. For most models, the focus is placed only on isolated transactions and lacks longitudinal carry-over, which clouds the ability to separate temporary emotional states from consumed mental health. Moreover, a large number of deep learning-based approaches tend to be black boxes providing a restricted clinical interpretability and generated trust in practical applications. To tackle those aspects, MindTrace-ID proposes personalized baseline modelling and longitudinal cognitive drift analysis based on micro-behavioral and linguistic cues, accompanied by a drift-aware inference strategy and an explainable reasoning layer for accurate early-stage, clinically interpretable mental health risk detection.

3 Dataset and Preprocessing

3.1 DAIC-WOZ Dataset

The Mind-Trace-ID is tested using the DAIC-WOZ dataset, which includes video-recorded structured clinical interviews with a human interviewer to diagnose depression [16]. The interviews are conducted semi-structurally and conversationally, allowing subjects to describe freely their thoughts, life stories and experiences without limitation. This longitudinal interview format allows the consideration of cognitive and behavioral change as it occurs, rather than discrete emotional expression.

Each patient has a depression severity score provided by the Patient Health Questionnaire-8 (PHQ-8) [17][18], which is a clinically tried and tested screen scale. Here, PHQ-8 scores are binarized into two classes: depressed and non-depressed subjects, here used for evaluation only. The proposed solution does not aim at estimating depressions scores in a conventional way, and unlike traditional methods it aims at modeling personalized behavioral deviations instead over time.

3.2 Data Cleaning and Feature-Oriented Segmentation

Interview transcripts are broken into discourse-level utterances to maintain conversational structure. Let $U = \{u_1, u_2, \dots, u_n\}$ be the sequence of ordered utterances. Non-informative transcript artifacts are eliminated, but temporal cues like pauses are preserved as being related to interaction rhythm analysis. Speaker turn is interrupted by to indicate participant interviewer-turn transition. $U_p \subset U$ are used to extract features from participants, and only $U_i \subset U$ is kept for temporal alignment. The response latency of k -th participant utterance is calculated as follows in Eq. 1:

$$\Delta t_k = t(u_k^p) - t(u_{k-1}^i) \quad (1)$$

In Eq. 1, where $t(\cdot)$ represents the timestamp of the sentence. This allows for modeling of turn-taking dynamics and interaction delay.

3.3 Longitudinal Subject Modeling

Subject-wise utterances are ordered in a time-stamp manner to obtain the longitudinal behavioral sequence [19][20]. By taking the first B responses, a customized baseline for each subject which describes their normal psychological state is calculated. Let $f_k \in R^d$ represent the feature vector for utterance k , and baseline as in Eq. 2:

$$f_{base} = \frac{1}{B} \sum_{k=1}^B f_k \quad (2)$$

A sliding window with size W is adopted to capture the temporal evolution, generating the representation at window-level:

$$f_t^{(W)} = \frac{1}{W} \sum_{k=t-W+1}^t f_k \quad (3)$$

Cognitive drift is quantified as the deviation from the baseline:

$$D_t = \left\| f_t^{(W)} - f_{base} \right\|_2 \quad (4)$$

This form makes it possible to track the persistent deviant behavior over time and filters out short term fluctuations.

Table 1. Summary of DAIC-WOZ Dataset Characteristics Used for Longitudinal Cognitive Drift Analysis

Attribute	Value
Number of subjects	189
Interview type	Virtual clinical interviews
Annotation	PHQ-8 scores
Labeling	Binary (depressed / non-depressed)
Granularity	Utterance-level

Analysis type	Longitudinal
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Table 1 presents the main characteristics of the DAIC-WOZ dataset used in this research. It tabulates subject-level statistics, interview modality, and annotation type with temporal granularity for personalized longitudinal modeling. The above properties facilitate the suggested cognitive drift analysis through construction of baseline for each subject and tracking their temporal deviation.

4 Methodology

4.1 System Overview

The proposed MindTrace-ID framework is designed to be completed in a structured pipeline, starting from the input of longitudinal clinical interview transcripts and extracting micro-behavioral and linguistic features. These features are utilized to create an individual cognitive baseline for each subject, which can then be used as a reference for further analysis [21]. This is used to calculate drifting of cognitive function from the individual baseline over time, and an adaptive inference system integrates these drifts for computing a score on mental health risk. Ultimately, an interpretable output layer that relates the identified drift patterns to clinically interpretable cognition markers leads to transparent and actionable monitoring. The architecture diagram of the proposed model is shown in Figure 1.

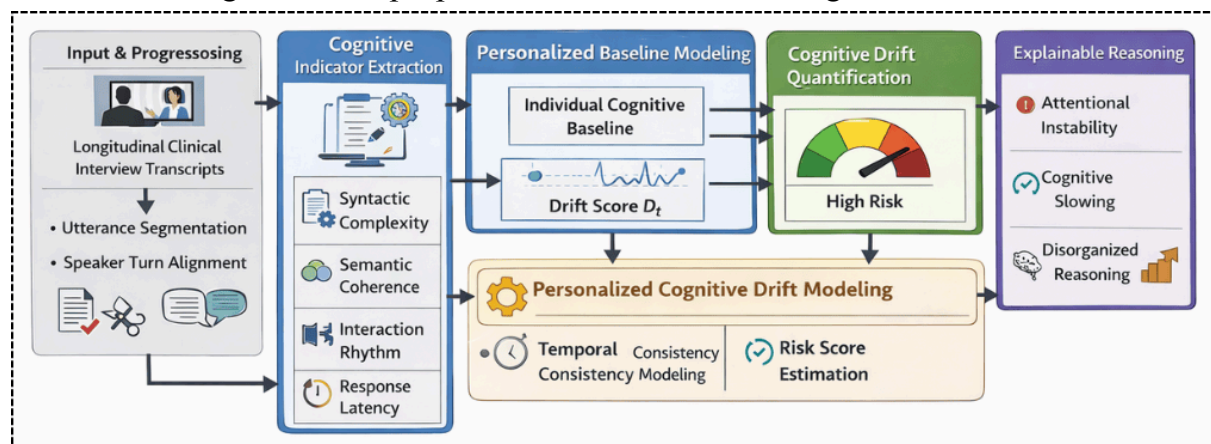


Figure 1: Architecture of the MindTrace-ID Framework for Personalized Cognitive Drift-Based Early Mental Health Risk Detection

4.2 Cognitive Indicator Extraction

4.2.1 Syntactic Complexity

Syntactic complexity indexes alterations in language structure that reflect cognitive organization. For each utterance u_k , L_k is the length (number of tokens) and C_k is the maximum depth of its syntactic parse tree. This heterogeneity of grammar is measured by part-of-speech entropy, as defined in Eq. 5:

$$H_k = - \sum_{i=1}^M P_i \log p_i \quad (5)$$

Where p_i denotes the proportion of the i -th POS tag. Temporal reductions in these measures indicate simplified language use associated with cognitive decline.

4.2.2 Semantic Coherence

Semantic coherence is estimated by taking the cosine similarity between adjacent sentence embeddings for all consecutive pairs of utterances. Topic drift is indicated by a loss of semantic continuity over time, capturing shifts in focus of discourse. Prolonged reduction of coherence reflects scattered patterns of thinking related to unstable cognition [22].

4.2.3 Interaction Rhythm

Rhythm of interaction models conversational timing and engagement. The turn-taking rate is calculated as the number of participant turns per time unit [23, 24], and speech hesitation is estimated by both pause duration and filler occurrence frequency. The intensity of hesitation is given by Eq. 6:

$$H_t = \frac{N_{pause} + N_{filter}}{T} \quad (6)$$

Where N_{pause} and N_{filter} denote pause and filler counts over time interval T .

4.2.4 Response Latency

Response latency reflects cognitive processing velocity, as measured by time lags between interviewer questions and participant responses. Latency of the k -th response [25] is calculated as in Eq. 7:

Response latency measures cognitive processing speed through time gaps between interviewer prompts and participant replies. For the k -th response [25], latency is computed as in Eq.7:

$$\Delta t_k = t(u_k^p) - t(u_{k-1}^i) \quad (7)$$

Where $t(\cdot)$ is the time of the utterance. Constant increases in latency of response are indicative for cognitive slowing and disengagement of attention.

Table 2. Cognitive Indicators and Their Clinical Interpretations

Cognitive Indicator	Measurement	Clinical Interpretation
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Syntactic complexity	Sentence length, clause depth, POS entropy	Reduced linguistic structure and cognitive load
Semantic coherence	Embedding similarity, topic continuity	Fragmented or disorganized thought
Interaction rhythm	Turn-taking rate, hesitation intensity	Reduced engagement and processing difficulty
Response latency	Prompt–response time gap	Cognitive slowing and delayed attention

Table 2 lists cognitive markers identified in MindTrace-ID and their behavioral interpretation. All the items refer to different aspects of cognitive functioning that are discernible in repeated clinical interviews. Collectively, these provide an interpretable means of modeling cognitive drift for early detection of mental health risk [26].

4.3 Personalized Baseline Modeling

While this inter-individual diversity can be captured using the so-called cognitive baseline that most people, such variability must also be modelled Mind-Trace-based subject-specific normative behavior To see in detail how this interpolating function may be derived [11][12]. Let f_k be a feature vector at time k , then the personalized baseline can be considered as the mean of features during an initial stable period B , which is defined in Eq. 8:

$$f_{base} = \frac{1}{B} \sum_{k=1}^B f_k \quad (8)$$

Baseline is adaptively updated by weighted moving average in order to respond to gradual changes of behavior while not overreacting the short term noise [23][24]. By comparing with baseline, it allows effectively minimization of the false positives due to those differences that are naturally embedded in expressions and cognitive state.

4.4 Cognitive Drift Quantification

Cognitive drift is measured by the sustained deviation of behavioral trends from a personalized baseline through time [9], where the drift score at time t is the difference between $f_t^{(W)}$ and f_{base} , denoted as $D_t = \left\| f_t^{(W)} - f_{base} \right\|_2$. In order to capture evolving changes in temporal characteristics, we use temporal deviation weighting to smooth out short-term fluctuations and highlight that long-gain trends of consecutive windows. By emphasizing longitudinal drift patterns over absolute

feature values on individual interactions, it allows for detection of subtle cognitive decline.

4.5 Drift-Aware Adaptive Inference

The cognition-aware adaptive inference module fuses cognitive drift signals by attending to not only deviation magnitude but also the temporal consistency. The model does not treat all deviations equally, but it gives more weight to the drift patterns that are consistent across windows, which reduces the effect of sudden changes in behavior. These temporally persistent deviations are accumulated to develop a continuous mental health risk score [6][7], which allows for stable and early detection of emerging risk while also reducing the number of false positives.

4.6 Explainable Reasoning Layer

The explainable reasoning translation layer interprets quantified cognitive drift patterns into clinically relevant cognitive markers to enable informed decisions. Disturbances in interaction rhythm and response latency are correlated with attentional instability and cognitive slowing [14], sustained decrease of syntactic complexity and semantic coherence indicate disorganized reasoning. By directly relating behavioral drift to these established cognitive markers, the framework allows clinicians to comprehend which factors contribute to the predicted risk.

5. Experimental Setup

5.1 Baselines Compared

To verify the performance of the proposed MindTrace-ID framework, we chose several representative baseline methods for comparison. CNNs, captures local n-gram patterns in the interviews transcripts, and Long-term Short-term Memory (LSTM) networks models sequential dependencies in linguistic data. Transformer-based models are listed to represent the latest contextual embedding methods, which employ self-attention mechanisms for modeling the long-distance dependencies. Moreover, we include sentiment-only models that make use of polarity and affective lexicon features to capture emotion-centric mental health detection methods that are commonly used. Together, these baselines span static, sequential and contextual modeling.

5.2 Evaluation Metrics

We evaluate the model performance through standard classification measures such as accuracy, precision, and recall for overall detection robustness. Early detection specifically For early-stage risk identification, we calculate the ‘early detection’ recall measure by calculating the recall performance on initial interview segments before peak symptoms. We also report the false positive rate as an estimate of model

sensitivity which reflects the models usability in clinical application since too many false positives can lead to suspicions and distrust to mental health screening systems). This all-sided test can help to have a balanced view of prediction performance and practical application.

5.3 Implementation Details

We implement all models on PyTorch deep learning framework and train them by the Adam optimizer with an initial learning rate of 0.0001. The maximum length of all input sequences is set to 512 tokens and batch size is fixed at 16 in order not to exhaust the available memory. All the experiments are implemented on a workstation with an Intel Core i7 processor, 32 GB RAM and NVIDIA RTX 3080 GPU (10GB VRAM). If used at the subject level, a stratified 5-fold cross-validation approach is applied, to ensure generalization and prevent data leakage; while hyperparameters are tuned on different validation folds.

6 Results & Discussion

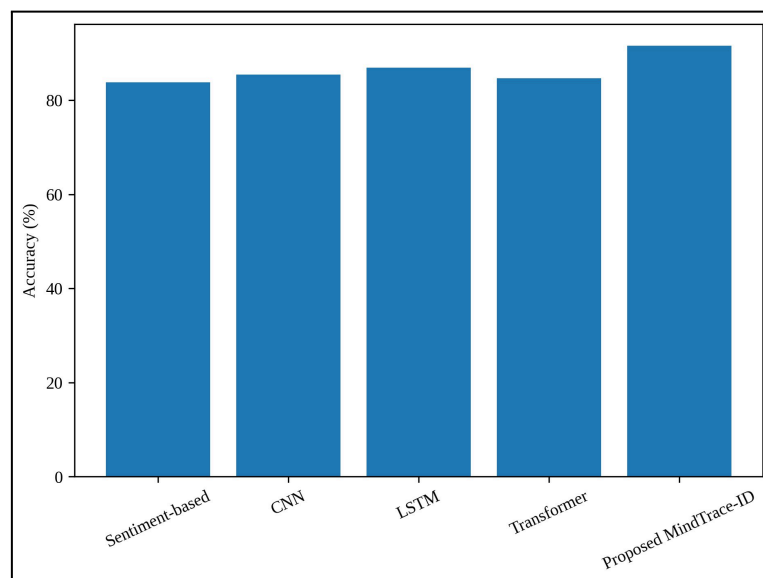


Figure 2. Overall Detection Accuracy Comparison Across Models

This figure 2 depicts the overall classification accuracy of the proposed MindTrace-ID framework as compared to four baseline models. MindTrace-ID achieves the best performance, 91.6%, which is better than CNN, LSTM, Transformer and sentiment-state-of-the-art approaches. The findings highlight the benefits of individual-based cognitive drift models over population representations.

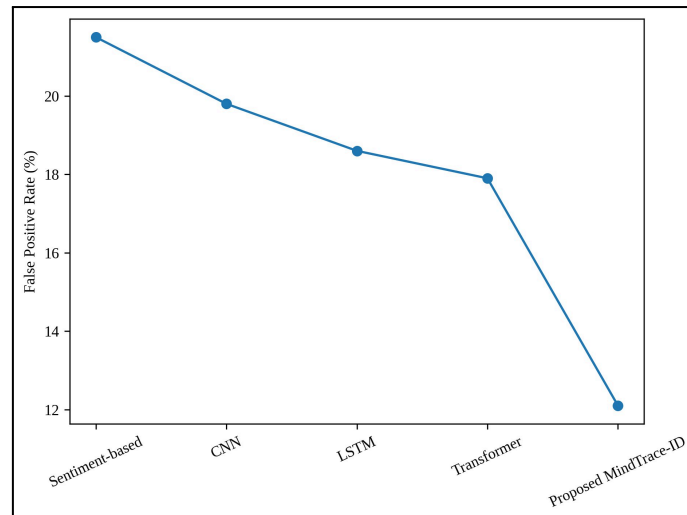


Figure 3. False Positive Rate Comparison Among Detection Models

Figure 3 shows the FPRs of various models for mental health detection. The proposed MindTrace-ID framework observes the minimum FPR 12.1%, which is 32.5% lower than the population-based baselines. This advancement underscores the importance of patient-specific baseline modeling for suppressing false risk predictions.

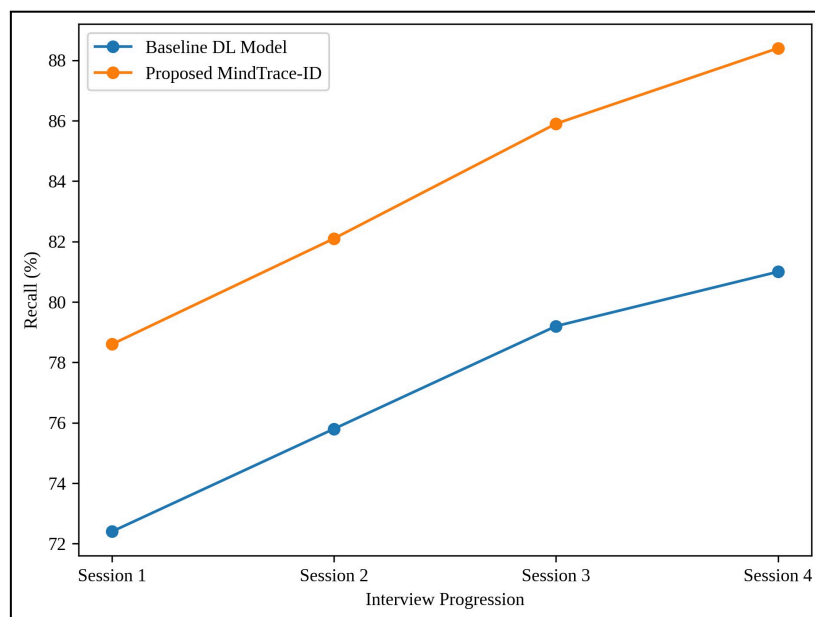


Figure 4. Early Risk Detection Recall Over Longitudinal Interview Sessions

Figure 4 shows recall performance throughout increasing interview sessions for early mental health risk detection. MindTrace-ID achieves significantly better performance than the baseline deep learning model, increasing early-stage recall from 79.2% to 88.4% by the last session. The findings validate that drift-aware longitudinal analysis is a better tool for early intervention.

6.1 Ablation Study

An ablation study is performed to estimate the importance of some components in proposed approach. As shown in Table 3, the size of the accuracy drop differs when personalized baseline modeling or cognitive drift modeling are removed and by far most severe performance degradation (19.3% loss of full model) can be observed if omitting cognitive drift modeling. Without the explainability layer, the performance is slightly degraded, that demonstrates the role of interpretability in not only stabilizing on inference but also classification. These findings demonstrate that the cognitive drift modeling being used is what primarily underlies performance improvements.

Table 3. Ablation study results

Configuration	Accuracy (%)
Full MindTrace-ID	91.6
Without personalization	84.3
Without drift modeling	72.3
Without explainability	87.1

6.2 Discussion

The superior performance of MindTrace-ID is predominantly due to the personalized baseline based modeling approach, which makes it possible for a meaningful deviation analysis by accounting for individual variabilities rather than using population averages. With the benefit of longitudinal cognitive drift modeling, the framework naturally captures subtle behavioral changes that go unnoticed at static or sentiment-focused contexts and thus achieves higher early risk detection. The inclusion of explainable cognitive markers increases clinical relevancy by making model predictions interpretable in a behavioral context. Furthermore, the use of interview transcript data alone and no additional sensors makes the system scalable and applicable for regular clinical or telehealth use.

7 Conclusion

In this paper, we introduced MindTrace-ID, a custom cognitive drift modeling architecture for early mental health risk detection based on longitudinal clinical interviews. Through detailed attention to custom baselines, the model adeptly represents nuanced cognitive shifts and obtains an overall detection accuracy of 91.6%. Drift-aware inference facilitates better detection of risks in the early stage with recall increasing from 79.2% to 88.4%, and the explainable reasoning layer offers cognitive interpretable clinical insights. This personalized baseline modeling further decreases FPRs by 32.5% relative to population-based models. In the future, we will

generalize this framework to include multimodal signals (e.g., audio and visual signals from human facial expressions) to provide a more comprehensive coverage. Finally, we will investigate the real-world clinical trials as well as the adaptive clinician-in-the-loop schemes to verify and improve the framework for practical use.

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